MILITARY SPECIFICATION

PROPELLING UNIT, OUTBOARD, DED

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a marine type, diesel-engine-driven, outboard propelling unit for propelling barges and pontoons.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-T-60 - Tape: Packaging, Waterproof.

MILITARY

MIL-C-104 - Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture and Fungus Resistant.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-L-2105 - Lubricating Oil, Gear, Multipurpose.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-3774 - Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-C-5501 - Cap and Plug, Protective, Dust and Moisture Seal.
MIL-E-10062 - Engines: Preparation for Shipment and Storage.
MIL-T-22085 - Tapes, Adhesive, Preservation and Sealing.

STANDARD

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

Gear Standards.

(Application for copies should be addressed to the American Gear Manufacturers Association, No. 1 Thomas Circle, N.W., Washington, DC 20005.)

NATIONAL LUBRICATING GREASE INSTITUTE (NLGI)

ASA Z11.130 - 1963 - Classification of Lubricating Grease.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

J534 - Standard for Lubrication fittings.
J755 - Standard for Marine Propeller-Shaft Ends and Hubs.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. The propelling unit shall consist of a raw water heat exchanger cooled marine diesel engine, outboard drive assembly with propeller, and related equipment, hydraulically powered propeller elevating mechanism(s), steel framework mounting assembly, integral control station and remote control.

3.2 First article. The contractor shall furnish a propelling unit for first article inspection and approval (see 4.2.1 and 6.2).

3.3 Engine approval and warranty certification. Prior to approval of the first article, if one is submitted or prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer from the diesel marine engine manufacturer that the diesel marine engine he proposes to supply under this specification meets the approval and warranty requirements of the manufacturer for marine application.

3.4 Standard commercial product. The propelling unit shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the propelling unit being furnished. A standard commercial product is a product which, has been sold or is currently being offered for sale, on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.5 Interchangeability. All propelling units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.7 Dissimilar metals. The use of dissimilar metals in intimate contact with each other shall be avoided to minimize galvanic corrosion. When the use of dissimilar metals cannot be avoided and injurious galvalic corrosion can be expected, an interposing material which will minimize or eliminate this effect shall be used.

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3.8 Performance. The engine shall start and accept rated thrust in not more than 5 minutes at ambient temperatures from 0 degrees Fahrenheit (°F) to 110°F. When the engine is operated at speed corresponding to the manufacturer's continuous duty rating, the bollard thrust shall be not less than 4,400 pounds. The rotatable outboard drive assembly shall complete one revolution within not less than 15 seconds. A shear pin, slip clutch or other approved suitable means shall be provided to prevent damage to the propeller drive mechanism in the event the propeller strikes an underwater obstacle with sufficient force to cause permanent damage to the drive train.

3.9 Design and construction. The propelling unit shall be designed to provide long periods of reliable all-weather service. Propeller drive shaft (stem) length shall be 60 inches nominal, length of drive shaft shall be measured from the deck of barge to centerline of propeller. All parts, components, and subassemblies, subject to service, repair, or replacement in the field shall be readily accessible, and conveniently secured with adequate protection from damage, parts and lubricant loss, and entrance of foreign liquid and material during continued rugged service. All moving and high temperature parts which may be hazardous to operating and maintenance personnel shall be enclosed or guarded. The design and materials of the mounting frames and supports shall be of sufficient strength to withstand the dead weight load of the complete propelling unit, plus engine vibrational forces, plus a 1.5 shock loading factor, plus navigational forces, without damage or permanent distortion of the frame. Mechanical gearing and powertrains shall be rated in accordance with AGMA Standards, at 150 percent of the input horsepower from the engine. Mechanical gearing and drives shall be fully enclosed in leakproof housings. Means shall be provided for filling, checking, and draining liquid lubricants and hydraulic fluid without disassembly. Outboard drive units, if disconnectable for shipping, shall be designed to prevent exposure of interior working parts and the loss of lubricants or hydraulic fluid. Hydraulic pumps, hydraulic motors, power hose, fittings, and controls shall conform to current SAE Standards, as applicable to the equipment and parts used. The essential subassemblies shall be engine with enclosure, heat exchanger, outboard drive, control station, electric starting system, hydraulic power distribution system, and related accesories. Engine, starter, heat exchanger, hydraulic reservoir, and related equipment shall be grouped with maximum practicable compactness on the deck mounting assembly framework (adjacent to the control panel), under a protective enclosure. Drain outlets for the engine coolant, crankcase oil, and hydraulic power distribution fluid shall be readily accessible for maintenance. The overall width of 2 propulsion units placed side by side on a barge in fully operational condition, utilizing remote controls specified herein, shall not exceed 12 feet. Individual propulsion units shall not exceed 12 feet in overall length including fuel and hydraulic tanks, measured from the middle of the leg when in the perpendicular position, to the forward most structural member of the unit. The overall height of the propelling unit complete and in operating mode shall not exceed 54 inches.

3.9.1 Engine monitoring control station. Engine monitoring control station shall be complete with conveniently arranged controls, and instrument panel. Controls shall be provided for one man operation. Electric cranking, charging system, and the instrumentation lighting system shall be an integral

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part of the units electrical system. All instrumentation shall be weatherproof. Instruments and controls shall include not less than the following:

a. Engine starting controls.
b. Manual clutch.
c. Steering control.
d. Outboard drive propeller elevation controls.
e. Instrumentation as specified in 3.9.3.
f. Remote control unit operable at not less than 4 feet from the propelling unit and propeller direction indicator. The remote control unit shall, as a minimum, have controls for steering and throttle setting.

3.9.1.1 Mechanical operated remote control unit. A mechanical operated remote control unit operable at not less than 4 feet from the propelling unit shall be provided. The control unit shall provide, as a minimum, the following:

a. Propeller direction indicator.
b. Mechanical controls for steering and engine control.

.9.2 Outboard drive assembly. The outboard drive assembly shall consist of an enclosed vertical drive shaft or hydraulic power hose assembly, rotatable through 360°, with a 90° angle gear drive, and a propeller. Hydraulic ram and piston control rods, when furnished, shall be made of suitable alloy, or heavily chromium plated, to preclude atmospheric corrosion or electrochemical decomposition. The propeller assembly shall be driven, either through drive shaft and bevel gearing or universal power transmission. The outboard drive assembly shall be capable of maintaining full power to the propeller when positioned from the perpendicular to a position where the top of the propeller is 12 inches below the plane of the base of the propulsion units. Gear ratios or hydraulic motor speeds, shall be selected to obtain propeller rotative speeds as required to provide the thrust as specified herein. The outboard drive assembly shall be provided with an oil lubrication system specifically designed for the application furnished. A convenient means of checking level of oil, draining, and filling shall be provided. Maintenance service shall not require disassembly. The outboard drive assembly shall be leakproof under all conditions of service. Disconnectable outboard drive assemblies shall be provided with oil seals and barriers to prevent loss of lubricant, or caps and plugs for hydraulic hose as applicable, when disassembled or shipped.

3.9.2.1 Outboard leg cockup. The outboard leg assembly shall have the ability to be cocked up hydraulically to not less than 90° from its normal operating (against the stops) to allow maintenance or repair function without docking.

3.9.2.2 Steering. The steerable vertical column shall be mounted in such a manner that the column and propeller assembly can be rotated a full 360° about the vertical axis. Full power of the propeller shall be available at any angle of steering rotation. Steering shall be accomplished by power actuated hydraulic motor with worm gear drive, and shall be controlled by a lever at the control station.

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3.9.2.3 Propellers. Propellers shall be of manganese bronze, stainless steel, or other commercially acceptable material and shall be bored and fitted with a suitable gasket, packing, or rope guard at the propeller shaft seal. Size and number of blades shall be as required for the applicable thrust. Propeller shaft ends and propeller hubs shall conform to SAE J755.

3.9.3 Diesel engine. The engine shall be a six cylinder, standard commercial marine type diesel engine with a closed cooling system. The engine shall have not less than 175 brake horsepower at 1800 revolutions per minute. The engine shall be furnished with the following accesories and equipment:

a. Full flow oil filter.
b. A fuel tank with capacity of not less than 90 gallons. The fuel tank shall be mounted either within the main frame under the engine or at either end of the frame and mounted directly to the main frame.
c. Closed heat exchanger system with the engine liquid coolant temperature thermostatically controlled.
d. Air cleaner or air silencer located to prevent the entrance of water.
e. High cooling liquid temperature warning device.
f. Low lubricating oil pressure warning device.
g. Manually operated, emergency overspeed shutoff control within convenient reach of the operator at the control station.
h. An instrument panel complete with a lubricating oil pressure gage; cooling liquid temperature indicator; mechanical tachometer; a mechanical type fuel gage mounted on the fuel tank, an ammeter, and an engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished to register accurately the number of hours of engine operating time. Meter shall be of strong rugged construction to insure continuous trouble free performance under the most extreme equipment operating conditions indicated in this specification.
i. Corrosion-resistant dry type muffler(s) suitable for marine duty. Exhaust outlets shall extend to the rear of the unit and shall not exceed the overall height requirements of the propelling unit (see 3.9).
j. Hydraulic power distribution system fluid pressure gage with normal pressure range indexed.
k. Outboard drive lubricating oil pressure gage with normal pressure range indexed.
l. Electric cranking, charging, and integral instrumentation lighting systems with a marine type battery(s) enclosed within a watertight enclosure.

3.9.4 Hydraulic systems. A complete hydraulic power distribution system shall be provided for steering control. A hydraulic filtration system shall be installed.

3.9.5 Electrical systems. The complete marine type electrical systems shall be provided for cranking, charging, and intrumentation lighting of the propelling unit. The systems shall incorporate a battery(s) for electrical

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storage and shall be enclosed within a watertight enclosure. All electrical components shall be of the type and quality utilized in good commercial marine applications for equipment of this type.

3.10 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 pound-force per square inch or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.11 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.12 Instruction plates. The propelling unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws, rivets, or bolts of not less than 1/8-inch diameter.

3.13 Identification plate. The contracting officer will furnish to the Government inspector, the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly and/or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8 inch in diameter shall be used to affix the plates.

3.14 Stenciling. The gross weight of each propelling unit shall be stenciled on each side of the unit. Stenciling shall conform to MIL-T-704.

3.15 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

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3.16 Workmanship.

3.16.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.16.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.16.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.16.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.16.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting ability to perform its intended function.

3.16.6 Assembled propelling unit. The assembled propelling unit shall be free from any defects that could cause malfunctioning of components or interfere with safe operation. Construction and applied coating shall be compatible with the best quality workmanship for this type of equipment.

3.16.7 Threaded fasteners. Threaded fasteners shall not be loose or missing. Threads shall not be missing, stripped, or broken and conform to FED-STD-H28.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable documents specified herein.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one propelling unit. This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be a first production item or a standard production item from the contractor's current inventory provided the propelling unit meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining propelling unit to be furnished under the contract, (see 6.2).

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.5.

4.3 Examination. Each propelling unit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. Permanent records, available to the Goverment, shall be kept of all test results and performance measurements. Production units shall be tested in accordance with 4.4.1. The first article shall be tested in accordance with 4.4.2. All defects shall be corrected. Noncorrectable defects shall be cause for rejection under the terms of the contract. After the completion of testing, the oil in the outboard drive of mechanical tranamissions shall be drained and examined for presence of water. Presence of water constitutes failure of the test. Lubricating oil pumps and lubricating system in mechanical outboard drives shall be bench tested during the initial assembly of all units. Loss of fluid in hydraulic gear motor drives shall be cause for rejection.

4.4.1 Production unit tests. Each propelling unit shall be tested for engine start-up, engine oil pressure, engine coolant temperature, steering and engine throttle control for 30 minutes. Not less than one propelling unit specified under each contract shall be tested for bollard thrust.

4.4.2 First article tests. The first article shall be mounted on a barge furnished by the contractor of sufficient size to adequately perform the testing required. The barge shall be loaded either by deck cargo or flooding to obtain a freeboard of between 10 and 15 inches. The propelling unit shall

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then be operated under field conditions with not less than two per hour each of tight turns and crash stops. First article testing shall be continued for not less than 16 hours of 6 hours (average) per day performance. Each day of scheduled operations shall include not less than the following:

a. All operations shall be conducted at maximum governed engine speed.
b. Four crash stops from maximum speed, accomplished by rotating the outboard drive 180° with the steering controls.
c. Check operating time on the engine hour meter and cruise at random until the equivalent of 6 engine hours at rated engine speed have been completed each day, for a total of not less than 16 hours of operation without breakdown.

Observations shall be made to determine prompt starting, steady running, response to controls, and the proper functioning of every item specified herein. Any malfunction shall constitute a failure of the test. The propelling unit shall be attached at a point in the plane of the centerline of the vertical outboard drive housing, to a stationary object, with a horizontal line of sufficient length to dissipate any backwash effect and incorporating a calibrated dynamometer. The outboard propelling unit shall be operated in gear at varying engine speeds up to full throttle. Engine speed versus bollard thrust (static pull) shall be recorded. Bollard thrust at the manufacturer's recommended engine speed for continuous duty shall be checked for conformance to the requirements of 3.8. Failure to attain this value shall constitute failure of the test.

4.4.3 First article sample. Upon acceptance of the article, the first article shall remain at the manufacturing facility as a production sample, and shall be the last unit delivered on the contract. The first article shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first article in a serviceable condition for the duration of the contract.

4.5 Packaging inspection. The preservation, packing, and marking of the propelling unit shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level B as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

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es of restraint harnesses, crash-worthy restraint systems, protect

5.1.1.3 Unit protection. The complete propelling unit shall be preserved method IIa. Openings in the equipment shall not be sealed, so that desiccated air may circulate.

5.1.1.4 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.5 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.6 Engines. Engines, engine components, and accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type V, method IIa, with the following exceptions:

a. The cooling system shall be processed by the water and antifreeze procedure.
b. The fuel system shall be processed using P-10, grade 10 preservative.

5.1.1.7 Hydraulic systems. The hydraulic fluid supply tanks shall be filled to the operating level with hydraulic fluid required for operation. A weatherproof tag shall be attached in a conspicuous location indicating: "The hydraulic supply tank is filled to the operating level with fluid required for operation. Do not drain". Markings shall be applied to the tag with a waterproof material.

5.1.1.8 Enclosed gears. Enclosed gears which operate in lubricants shall be preserved by having the housing filled to the operating level with one of the following applicable preservatives:

Operating Lubricant	Preservative Lubricant
Lubricating oil (SAE 10, 30, or 50)	P-10, grade as applicable
Gear lubricant (SAE 75, 80, 85, 90, or 140)	MIL-L-2105, grade as applicable
Lubricant other than above	Manufacturer's approved operating lubricant

Gear mechanism shall be operated to insure coating of all interior surfaces with preservative. Gear housings containing any of the above lubricants shall be identified with a weatherproof tag to indicate: "This housing is filled to the operating level with lubricant. Do not drain until first required lubrication change". Markings shall be applied to the tags with a waterproof material. The tags shall be attached in a conspicuous location.

5.1.1.9 Pumps. Interior surfaces of pumps shall be coated with type P-10, grade 30 preservative in a manner to insure thorough coating of all interior parts and surfaces. Excess preservative shall be drained.

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5.1.1.10 Grease lubrication. Bearings and other components requiring a multipurpose, extreme pressure (EP), or NLGI No. 2 grease for operational use shall be preserved with type P-11 preservative.

5.1.1.11 Propellers. Unpainted propellers shall be preserved with type P-1 preservative.

5.1.2 Level B.

5.1.2.1 Unit protection. The complete propelling unit shall be preserved method I. Preservation shall be the same as level A with the following exceptions:

5.1.2.2 Engines. Engines, engine components, and accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type V, method I with the following exceptions:

a. The cooling system shall be processed by the water and antifreeze procedure.
b. The fuel system shall be processed using P-10, grade 10 preservative.

5.1.2.3 Hydraulic systems. The hydraulic fluid supply tanks shall be filled to the operating level with hydraulic fluid required for operation. The pistons shall be retracted as far as practicable into the cylinders and secured. When the pistons cannot be fully retracted, the exposed portions of the piston rods (ramshafts) shall be coated with type P-6 or P-11 preservative and the coated surfaces wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, extending the wraps approximately 2 inches onto the ram cylinders. The wraps shall be secured in place with waterproof tape. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position and preserved as specified herein for piston rods. Hoses shall not be disconnected. A weatherproof tag shall be attached to the control lever indicating: "The hydraulic supply tank is filled to the operating level with fluid required for operation. Do not drain." Markings shall be applied to the tag with a waterproof material.

5.1.2.4 Pumps. Interior surfaces of pumps shall be coated with type P-10, grade preservative in a manner to insure thorough coating of all interior parts and surfaces. Excess preservative shall be drained. All openings into the pumps shall be sealed with caps or plugs conforming to MIL-C-5501 or waterproof tape conforming to MIL-T-22085, type II, or PPP-T-60, type IV.

5.1.3 Level C. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage. The equipment shall be lubricated for operational service as required in the operator's manual.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

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5.2.1 Level A. Each propelling unit shall be packed in a crate conforming to MIL-C-104, type II, class and style optional. Anchoring, cushioning, blocking, and bracing of the contents shall be in accordance with the appendix to MIL-C-104.

5.2.2 Level C. Each propelling unit shall be packed in a crate conforming to MIL-C-3774, type II, style optional. Anchoring, cushioning, blocking, bracing, and waterproofing with a half shroud in accordance with the appendix to MIL-C-3774.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The propelling unit covered by this specification is intended for use at established Naval Stations for propelling barges and pontoons.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval (see 3.2, 4.2.1 and 6.4).
c. When the color of the finish coat is specified (see 3.11).
d. When fungus resistance is required (see 3.15).
e. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

Preparing Activity:

Navy - YD

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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